

Zayd Krunz

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Education	BASIS Tucson North	High School • 3.86 GPA (4.62 weighted)
	Tucson, AZ	August 2023 - May 2027
	• #1 Ranked High School in the Nation (<i>U.S. News & World Report</i> , 2025-2026)	
	• SAT: 1570, National Merit Semifinalist, USACO Silver, Codeforces Specialist (1480)	

Experience	LOCH Technologies	Emeryville, CA
	Intern (paid)	July 2026
	• Architected an ML-based User Equipment (UE) detection framework for the AirShield product, utilizing Software-Defined Radios (SDR) to identify unauthorized mobile devices in highly regulated environments.	
	• Trained and validated predictive machine learning models for RF signal classification, engineering the pipeline to ingest, process, and evaluate synthetic SDR datasets generated in MATLAB.	
	https://loch.io	
	SmartNet Communications	Tucson, AZ
	Freelance Web Developer	September 2024 - November 2024
	• Designed and deployed a responsive single-page application for a client using Next.js, improving site load metrics by over 40% via static site generation (SSG).	
	https://smartnetcommunication.com	

Research & Programs	University of Arizona & NSF WISPER Center	December 2025 - January 2026
	• Co-authored a technical poster "Weakly Supervised Machine Learning for Enhancing RFI Detection in Passive/Active Coexistence Scenarios" presented at the NSF WISPER Workshop (Jan 2026).	
	• Collaborated with PhD candidates to implement Semi-Supervised GANs (SGAN) and InfoGAN architectures to mitigate Radio Frequency Interference (RFI) in sub-THz astronomical bands.	
	https://zaydkrunz.com/research	
	NSF AI-EDGE Institute (Undergraduate Research)	June 2025 - July 2025
	• Evaluated performance trade-offs between Vision Transformer (ViT) and Convolutional Neural Network (CNN) models by benchmarking accuracy against training epochs and steps.	
	• Compiled and analyzed the nanoGPT codebase to deconstruct the foundational architecture of a transformer-based language model.	
	https://aiedge.osu.edu	
	Stanford Pre-Collegiate Studies	July 2025
	• Engineered a predictive maintenance model by evaluating 12 algorithms (Random Forest, XGBoost, etc.) to detect rare equipment failures.	
	• Deployed a voting classifier optimized for high recall to minimize costly downtime events.	
	https://zaydkrunz.com/spcs	

Skills	Programming Languages	
	C++, Python, TypeScript, JavaScript, SQL, HTML/CSS, Lua, LaTeX	
	Frameworks & Libraries	
	PyTorch, scikit-learn, NumPy, Pandas, Next.js, React, Tailwind CSS, Prisma, Bun	
	Cloud & Tools	
	AWS, GCP, Docker, Linux/Bash, PostgreSQL, Cloudflare, Git	

Languages	Trilingual (Native Proficiency in English, French, and Arabic)
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